1 Introduction

Standardized testing has always played an important role in the process of college admission in the United States. Standardized tests such as the SAT have been incorporated in order to assess what level of preparedness a student has academically, which in turn allows colleges to compare students from a myriad of different backgrounds in a standardized fashion. This is important because students are coming from a multitude of different high schools from all across the United States. However, what role it plays in the process of being accepted into a college is becoming increasingly controversial in recent years. There are a number of reasons why this has become a controversy, one being that it may be a determining factor in what a student's socioeconomic status is as opposed to what level of preparedness a student demonstrates. It has always been recognized that the socioeconomic status of a student plays an important role in the level of success that a student exhibits in a number of different academic situations. For example, a student from a more affluent background has the opportunity to be exposed to a number of different academic tools that can help the student achieve success, such as tutoring opportunities and academically enriched learning environments. However, a student from a less affluent background may be subjected to a number of underlying factors that detract from the student's success in a number of different situations, such as the quality of school systems and learning opportunities that the student may be exposed to. These types of factors have a significant impact on a student's success in a variety of different situations, such as before a student is exposed to a variety of different standardized testing situations. It has long been understood that socioeconomic status and a student's level of success are inextricably linked. A study completed by Sirin (2005) involved a meta-analysis of studies that were completed in order

to examine the relationship between socioeconomic status and student achievement. This study clearly demonstrates a positive relationship between these two factors. It has been observed that a student whose socioeconomic status is classified as being at a higher level will have a better level of success in a variety of different areas, such as a standardized test. Another study completed by Reardon (2013) observed a gap in academic performance between high and low-income students in the United States. This demonstrates that economic inequality is a factor that may be able to impact a student's level of success in a variety of different ways. Another study completed on this type of testing also observed a correlation between socioeconomic status and SAT performance. Sackett et al. (2012) completed a study that examined the correlation between socioeconomic status and SAT performance in a variety of different ways. This study observed that socioeconomic status is a factor that may be able to impact a student's level of success in a variety of different ways, such as in a standardized test and in a student's SAT results in a college admissions process. This project is different from other studies in that it does not examine a student's level of success in this fashion; instead, it looks at a variety of different aggregate statistics in order to examine a variety of different trends throughout the nation. This project also hopes to observe whether or not there is a correlation between SAT performance and geographic regions. This allows it to explore a number of trends, which may help in building an understanding about the relationship between socioeconomic status and SAT performance.

2 Related Work

A lot of research has been carried out to assess the relationship between socioeconomic status and academic performance. It has been identified that students coming from higher socioeconomic status families tend to perform better in tests compared to those coming from

lower socioeconomic status families. Sirin (2005) conducted an extensive research by using the meta-analysis technique to examine the relationship between socioeconomic status and academic performance. Through the meta-analysis of a large number of research studies, Sirin has found a strong relationship between socioeconomic status and academic performance. Students coming from families with high socioeconomic status have been seen to have better academic performance in different academic domains. This indicates that socioeconomic factors have an important role to play in academic performance. A number of research studies have been conducted to find the relationship between socioeconomic factors and test performance. A research study was conducted by Sackett et al. in the year 2012 to find the relationship between socioeconomic status and SAT test performance. They found that socioeconomic factors play an important role in standardized tests as well as their interpretation in college admissions. This indicates that there is a strong relationship between socioeconomic factors and academic performance, which cannot be ignored in college admissions. Some studies have also been conducted to understand the educational inequality among students of different socioeconomic statuses. Reardon (2013) carried out a study and observed that the difference in the academic performance of students belonging to high and low income families is widening. This indicates that socioeconomic factors are playing an important role in the academic performance of students in the United States. This project is conducted with the objective of finding the relationship between income and SAT results at an institutional level. Instead of finding the relationship at the individual level, this project is conducted with the objective of finding the relationship between income and SAT results at the institutional level.

3 Data

The dataset used for this project consists of information regarding institutions and universities in the United States. This dataset provides information regarding the academic and socioeconomic aspects of students. Using a dataset based on institutions is helpful in finding large-scale trends because it aggregates data from a large number of individuals attending various institutions. There were three main variables in this analysis. The first variable is the average SAT score of students admitted into each institution. This is a main variable in this analysis and represents the overall performance of students attending that institution in a standardized test. The second main variable is the median income level of families whose students attend a particular institution. This gives a clear picture about the income level of people whose students are attending a particular institution and allows us to conduct a trend analysis about whether income levels have a significant impact on the overall performance in a standardized test. The third main variable in this dataset is the geographic location in which each institution is situated in the United States. Each institution in this dataset is classified based on the geographic location in which it is situated in the country. This allows us to conduct a trend analysis about whether there are significant differences in SAT test performances in different regions in the country. Before conducting this analysis, it is important to clean and preprocess this dataset. In this case, observations in which values were missing in some of the main variables such as SAT scores and income levels were discarded to ensure that the analysis is reliable and accurate. Data cleaning is a very important step in any data analysis project because it ensures that the results obtained from conducting such analysis are accurate and reliable.

4 Methodology

In this project, exploratory data analysis and statistical modeling techniques are used to examine the relationship between institutional median family income and SAT performance. Exploratory data analysis gives a visual overview of the relationships in the data, while statistical modeling allows the analysis to quantify these relationships. The first step in this analysis is to create visualizations to explore possible relationships between median family income and SAT performance. A scatter plot is created to visually examine the relationships between income levels at an institution and the average SAT score. A scatter plot is helpful in visualizing a relationship between two numerical variables. Each point in a scatter plot represents an institution and has a median family income as its x-coordinate and average SAT score as its y-coordinate. This allows patterns in the data to emerge. Additionally, to explore possible regional differences in SAT performance, box plots are created to compare SAT performance across geographic regions. Box plots are useful for visualizing the distribution of values across groups and for comparing values across groups. Once the exploratory analysis is complete, a linear regression analysis is performed to evaluate the relationship between median family income and SAT performance. In this analysis, SAT_AVG is used as the dependent variable and MD_FAMINC as the independent variable. This analysis allows the relationships in the data to be quantified. The analysis is performed using Python as the programming language. The pandas library is used for data manipulation and preparation for analysis. Matplotlib and seaborn libraries are used for visualization. These libraries are widely used tools for analyzing and visualizing complex datasets.

5 Results & Findings

The results of the exploratory data analysis revealed a noticeable association between the institutional median family incomes and average SAT scores. The visualization of the data, as revealed by the scatterplot, showed an upward trend of the data points. This meant that the institutions with higher median family incomes had higher average SAT scores. Although the data points did not show a perfectly linear trend, the distribution of the data points revealed a positive association between the levels of median family incomes and average SAT scores. This is because, as revealed by Figure 1, the visualization of the data points showed that the institutions with higher median family incomes had higher average SAT scores. Apart from the general upward trend of the data points, as revealed by the scatterplot, the distribution of the data points also revealed a positive association between the levels of median family incomes and average SAT scores. For instance, the data points revealed that the institutions with relatively lower levels of median family incomes tended to cluster more frequently in the lower range of SAT scores, while the data points also revealed that the institutions with higher levels of median family incomes tended to cluster more frequently in the higher range of SAT scores. This indicates that income levels may have a bearing on the overall academic preparedness of students in these institutions. Though this trend is not entirely consistent across all institutions, it may be noted that overall, income does have a bearing on differences in academic achievement. It must also be noted that there is a degree of variance in institutions that have similar income levels. Some institutions with relatively moderate income levels have high averages in SAT scores, while others with higher income levels have relatively moderate SAT averages. This may be seen to suggest that income levels are not entirely responsible for determining SAT scores. Other

factors may be at work in determining differences in SAT averages across institutions. However, it may be noted that the positive trend between income levels and SAT averages is evident in the overall pattern. In addition to the analysis of the levels of median family income and average SAT scores, the analysis also attempted to establish whether there were any regional differences in SAT scores. The box plot visualization of the data points indicated that there were variations in the distribution of SAT scores in the different regions of the world. Some regions had higher median scores in the SAT, while in other regions there were variations in the range of SAT scores. Figure 2 shows the box plot visualization of the data points, which indicated variations in the SAT scores in the regions of the world. The analysis of the data in the regions of the world indicated variations in the range of SAT scores in the regions of the world. Some regions had narrow distributions of SAT scores, indicating that the institutions in the regions had relatively similar academic performance, while in other regions there were variations in the range of SAT scores, indicating variations in the academic performance of the institutions in the regions. To further establish the relationship between median family income and average SAT scores, a regression analysis was performed. The results of the regression analysis showed a positive relationship between the levels of median family income and average SAT scores. As can be seen in Figure 3, the results of the regression analysis indicate a positive correlation with regard to the levels of median family income and average SAT scores. The regression model also confirms the results of the scatter plot, as it indicates a positive correlation with regard to the level of median family income and average SAT scores. Although the regression line provides a general overview of the correlation between the two variables, the scatter of points around the regression line also indicates that other variables may affect SAT scores. These variables may include

school quality, student preparation, and other educational systems. However, it is clear that there is a positive correlation with regard to the level of institutional income and SAT scores.

6 Discussions & Interpretations

The results of this analysis show the potential impact that socioeconomic status can have on the results of a standardized testing program. By examining the correlation between the median income level of the families in the testing group and the scores they received on the SAT, it is easy to see that the individuals with the higher socioeconomic status are the ones with the potential advantage in terms of the academic preparation they received in order to score well on a standardized test. At the same time, the results of the exploratory data analysis and regression modeling serve as an example of the benefits of using visual data analysis in detecting patterns in data, specifically in the sphere of education. The use of visual data analysis tools, such as scatterplots and boxplots, enables the researcher to detect patterns in the data that may not be immediately obvious based on the data alone. The use of regression analysis, as a modeling tool, enables the researcher to build a better understanding of the relationship between socioeconomic factors and academic outcomes. There are a number of limitations that must be taken into consideration in relation to the overall analysis and results of this specific study. One of the considerations in this case is the fact that the data set used in the analysis in this project is based on averages and does not take into account the individual student's performance. This means that the analysis in this project is not able to assess how an individual student may perform in relation to their own socioeconomic status. Another consideration is the fact that the regression analysis carried out in this project is not complex and does not take into account any other variables that

may be of significance in the analysis in an attempt to assess the complex factors that may affect the results of the SAT test. Overall, the analysis and results of this specific project demonstrate the potential for discussions in relation to educational inequality. The patterns that have been identified in relation to income levels and standardized test performance help to demonstrate the ways in which socioeconomic status may have an influence on academic opportunities.

7 Conclusions & Future Work

This project explored whether there is a relationship between median income levels and average SAT scores in colleges and universities throughout the United States. Data analysis and regression were used in the exploration of this phenomenon, showing that there is a positive relationship between income levels and the average SAT scores in our colleges and universities across the nation. While the purpose of this study was not to determine whether there is a cause-and-effect relationship between income levels and the average SAT scores, the purpose was to determine whether there was a potential relationship between socioeconomic status and testing outcomes, much like the findings of previous studies that have indicated a significant relationship between socioeconomic status and educational outcomes. This study shows that future studies could be done by extending the current study to include more variables, thus using more complex data analysis techniques to explore more issues in one study. Future studies could also be conducted by expanding this study to explore the relationship between socioeconomic status and testing outcomes further. This study reveals the potential that data analysis techniques have to help us understand many issues within our educational system, thus gaining a better understanding of our society and how it is structured. By exploring the relationship between

socioeconomic status and testing outcomes within our educational system, we can gain a better understanding of our society.

8 Figures

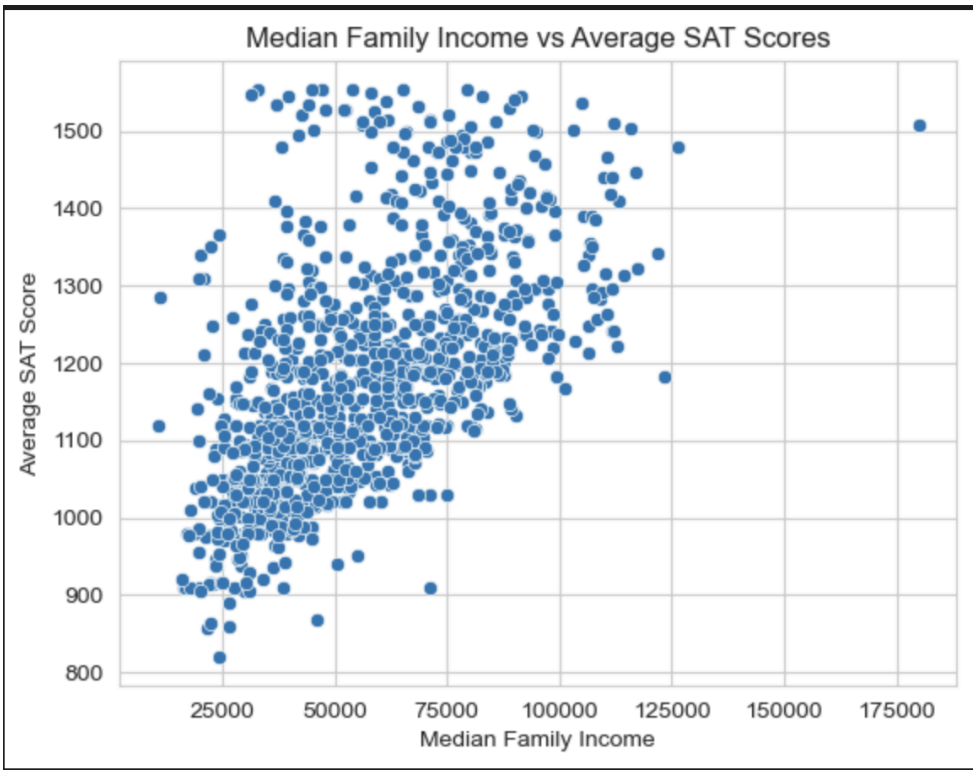

Figure 1. Relationship between median family income and average SAT scores across institutions. The scatterplot displays the association between institutional median family income and average SAT scores. Each point represents an institution, with higher income levels generally corresponding to higher SAT averages.

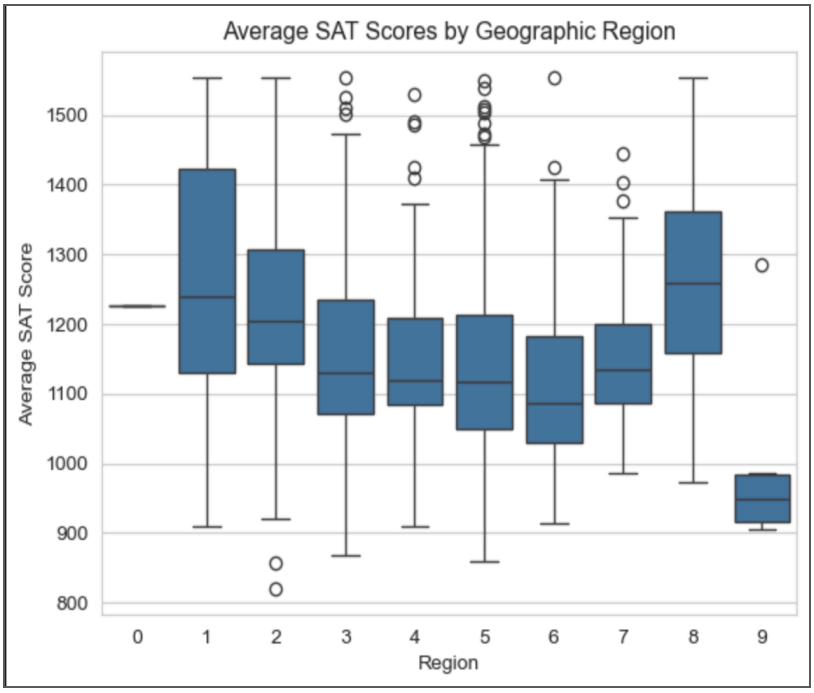

Figure 2. Distribution of SAT scores across geographic regions. This boxplot compares institutional SAT score distributions across U.S. regions. Differences in median values and score ranges illustrate variation in standardized testing outcomes across geographic contexts.

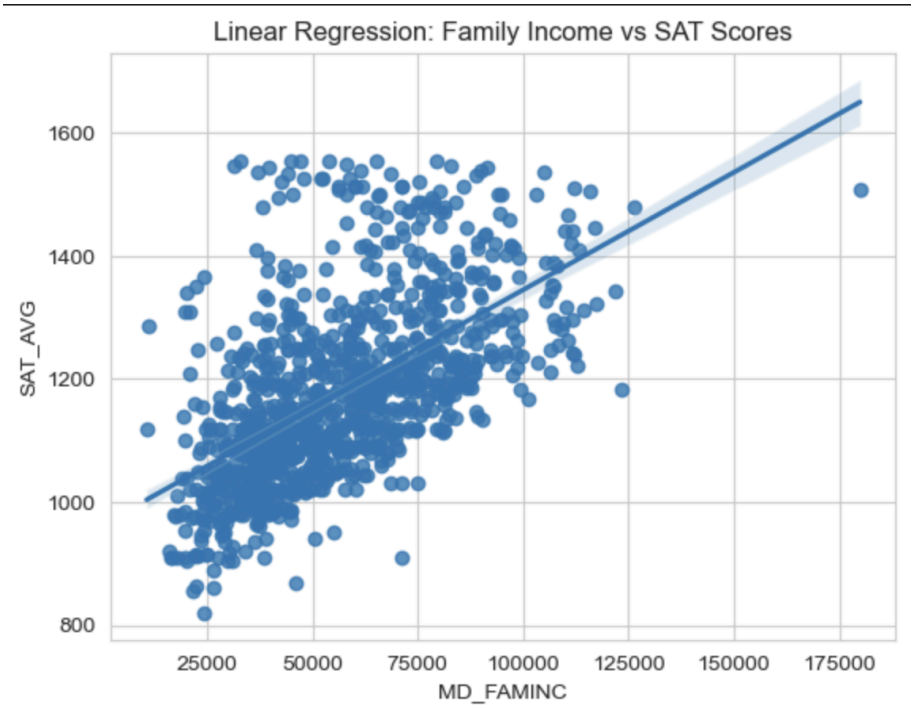

Figure 3. Linear regression model estimating the relationship between median family income and SAT scores. The regression line illustrates the positive association between institutional median family income and average SAT scores, supporting the pattern observed in the exploratory analysis.

Refrences

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